

1. A small ball of mass M and density ρ is dropped in a viscous liquid of density ρ_0 . After some time, the ball falls with a constant velocity. What is the viscous force on the ball?

[2023 (06 Apr Shift 1)]

- (1) $F = Mg\left(1 + \frac{\rho_0}{\rho}\right)$
- (2) $F = Mg\left(1 + \frac{\rho}{\rho_0}\right)$
- (3) $F = Mg\left(1 - \frac{\rho_0}{\rho}\right)$
- (4) $F = Mg(1 \pm \rho\rho_0)$

2. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: When you squeeze one end of a tube to get toothpaste out from the other end, Pascal's principle is observed.

Reason R: A change in the pressure applied to an enclosed incompressible fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.

In the light of the above statements, choose the most appropriate answer from the options given below.

[2023 (06 Apr Shift 2)]

- (1) Both A and R are correct but R is NOT the correct explanation of A
 - (2) A is not correct but R is correct
 - (3) A is correct but R is not correct
 - (4) Both A and R are correct and R is the correct explanation of A
3. An air bubble of volume 1 cm^3 rises from the bottom of a lake 40 m deep to the surface at a temperature of 12°C . The atmospheric pressure is $1 \times 10^5 \text{ Pa}$, the density of water is 1000 kg m^{-3} and $g = 10 \text{ m s}^{-2}$. There is no difference of the temperature of water at the depth of 40 m and on the surface. The volume of air bubble when it reaches the surface will be

[2023 (08 Apr Shift 1)]

- (1) 2 cm^3
- (2) 3 cm^3
- (3) 4 cm^3
- (4) 5 cm^3

4. An air bubble of diameter 6 mm rises steadily through a solution of density 1750 kg m^{-3} at the rate of 0.35 cm s^{-1} . The co-efficient of viscosity of the solution (neglect density of air) is _____ Pas (given, $g = 10 \text{ m s}^{-2}$).

[2023 (08 Apr Shift 1)]

5. A hydraulic automobile lift is designed to lift vehicles of mass 5000 kg . The area of cross section of the cylinder carrying load is 250 cm^2 . The maximum pressure the smaller piston would have to bear is [Assume $g = 10 \text{ m s}^{-2}$]

[2023 (08 Apr Shift 2)]

- (1) $20 \times 10^6 \text{ Pa}$
- (2) $2 \times 10^5 \text{ Pa}$
- (3) $200 \times 10^6 \text{ Pa}$
- (4) $2 \times 10^6 \text{ Pa}$

6. Given below are two statements:

Statement I: Pressure in a reservoir of water is same at all points at the same level of water.

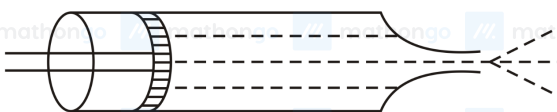
Statement II: The pressure applied to enclosed water is transmitted in all directions equally.

In the light of the above statements, choose the correct answer from the options given below:

[2023 (10 Apr Shift 1)]

- (1) Both Statement I and Statement II are false
- (2) Statement I is true but Statement II is false
- (3) Statement I is false but Statement II is true
- (4) Both Statement I and Statement II are true

7. Figure below shows a liquid being pushed out of the tube by a piston having area of cross section 2.0 cm^2 . The area of cross section at the outlet is 10 mm^2 . If the piston is pushed at a speed of 4 cm s^{-1} , the speed of outgoing fluid is _____ cm s^{-1}



[2023 (10 Apr Shift 2)]

9. The surface tension of soap solution is $3.5 \times 10^{-2} \text{ N m}^{-1}$. The amount of work done required to increase the radius of soap bubble from 10 cm to 20 cm is $\text{---} \times 10^{-4} \text{ J}$. (take $\pi = \frac{22}{7}$)

12.



ANSWER KEYS

1. (3) 2. (4) 3. (4) 4. (10) 5. (4) 6. (4) 7. (80) 8. (2)
9. (264) 10. (160) 11. (4) 12. (3) 13. (3) 14. (1150)

1. (3)

The buoyant force is given by

$$B = V\rho_0g$$

where V is the volume of the ball. The weight of the ball

$W = Mg = V\rho g$, is acting in the opposite direction of the buoyant force. Hence, at terminal velocity

$$Mg - B = F_v$$

$$\Rightarrow B + F_v = Mg$$

$$\Rightarrow F_v = V\rho g - V\rho_0g$$

$$\Rightarrow F_v = \rho Vg \left(1 - \frac{\rho_0}{\rho}\right)$$

$$\Rightarrow F_v = Mg \left(1 - \frac{\rho_0}{\rho}\right)$$

2. (4)

Pascal's law states that the pressure in a closed incompressible fluid is transmitted equally in all directions.

The formula for pressure is given by

$$P = \frac{F}{A}$$

When a toothpaste is squeezed let the pressure transmitted by P . At the mouth of the toothpaste the pressure will also be equal to P due to Pascal's law. The pressure applied to the tube is transmitted equally throughout the toothpaste. When the pressure reaches the open end, it forces toothpaste out through the opening. Hence, both A & R is correct and R is correct explanation for A .

3. (4)

The pressure at the bottom of the lake can be calculated as follows:

$$\begin{aligned} P_{\text{bottom}} &= P_0 + \rho gh \\ &= (10^5 + 1000 \times 10 \times 40) \text{ Pa} \\ &= 5 \times 10^5 \text{ Pa} \end{aligned}$$

As the temperature of water remains constant, it can be written that

$$P_{\text{bottom}} V_{\text{bottom}} = P_{\text{top}} V_{\text{top}} \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the required volume of the bubble when it reaches the surface.

$$\begin{aligned} 5 \times 10^5 \times 1 &= 1 \times 10^5 \times V_{\text{top}} \\ \Rightarrow V_{\text{top}} &= \frac{5 \times 10^5 \times 1}{1 \times 10^5} \text{ cm}^3 \\ &= 5 \text{ cm}^3 \end{aligned}$$

4. (10)

The bubble is moving with a constant velocity so the net force acting on the body is zero.

Stokes' law defines the drag force which is given by $F = 6\pi\eta r v$

$$\text{The buoyant force is given by } B = V\rho g = \frac{4}{3}\pi r^3 \rho g$$

Hence, it can be written that

$$\frac{4}{3}\pi r^3 \rho g = 6\pi\eta r v$$

$$\Rightarrow \eta = \frac{2r^2 \rho g}{9v} = \frac{2 \times (3 \times 10^{-3})^2 \times 1750 \times 10}{9 \times 0.35 \times 10^{-2}} = 10 \text{ Pas}$$

5. (4)

The given data is

$$m = 5000 \text{ kg}$$

$$g = 10 \text{ m s}^{-2}$$

$$A = 250 \times 10^{-4} \text{ m}^2$$

Using the formula of pressure,

$$P = \frac{F}{A} = \frac{mg}{A}$$

$$= \frac{5000 \times 10}{250 \times 10^{-4}}$$

$$= 2 \times 10^6 \text{ N m}^{-2}$$

$$= 2 \times 10^6 \text{ Pa}$$

6. (4)

Pressure in a static fluid will be same at each point on same horizontal level.

$$P = P_0 + \rho gh$$

where P_0 is atmospheric pressure.

Pascal's law states that pressure applied to enclosed water is transmitted equally in all directions. Hence, both the statements are true and this is the right option.

7. (80)

The given data is

$$A_1 = 2 \times 10^{-4} \text{ m}^2$$

$$A_2 = 10 \times 10^{-6} \text{ m}^2$$

$$v_1 = 4 \text{ cm s}^{-1}$$

Using equation of continuity which is a conservation of mass principle,

$$A_1 v_1 = A_2 v_2$$

$$\Rightarrow v_2 = \frac{A_1 v_1}{A_2} = \frac{2 \times 10^{-4} \times 4}{10 \times 10^{-6}}$$

$$= 80 \text{ cm s}^{-1}$$

8. (2)

The terminal velocity can be given by

$$v_T = \frac{2(\rho - \sigma)r^2g}{9\eta}$$

Clearly, $v_T \propto r^2$

Let the radius of the small drops be r_1 and the radius of the big drop be r_2 .

So,

$$8\left(\frac{4\pi}{3}\right)r_1^3 = \left(\frac{4\pi}{3}\right)r_2^3$$

$$\Rightarrow r_1 = \frac{r_2}{2}$$

Hence, taking the ratio

$$\frac{(v_T)_1}{(v_T)_2} = \left(\frac{r_1}{r_2}\right)^2$$

$$\Rightarrow \frac{10}{(v_T)_2} = \frac{1}{4}$$

$$\Rightarrow (v_T)_2 = 40 \text{ cm s}^{-1}$$

9. (264)

The amount of work done is equal to the change in the surface energy to increase the radius of the bubble.

The formula to calculate the change in surface energy is given by

$$\Delta U = 2(T \times \Delta A) \quad \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the required work done.

$$\Delta U = 3.5 \times 10^{-2} \times 4\pi \left[(0.2)^2 - (0.1)^2 \right] \times 2 \text{ J}$$

$$= 3.5 \times 10^{-2} \times 4 \times \frac{22}{7} \times 0.03 \times 2$$

$$= 2 \times 1.32 \times 10^{-2}$$

$$= 264 \times 10^{-4} \text{ J}$$

10. (160)

The formula to calculate the potential (V) of each tiny drop is given by

$$V = k \frac{q}{r} \dots (1)$$

When all the drops forms the bigger drop the radius of the new drop (r') can be calculated as follows:

Volume of big drop = $64 \times$ volume of each tiny drop

$$\Rightarrow \frac{4}{3} \pi r'^3 = 64 \times \frac{4}{3} \pi r^3$$

$$\Rightarrow r' = 4r$$

So, the potential of the new big drop (V') can be written as

$$V' = k \frac{64q}{r'}$$

$$= k \frac{64q}{4r}$$

$$= 16k \frac{q}{r} \dots (2)$$

Divide equation (2) by equation (1) and simplify to obtain the potential of the new drop.

$$\frac{V'}{V} = \frac{16k \frac{q}{r}}{k \frac{q}{r}}$$

$$\Rightarrow V' = 16V \dots (3)$$

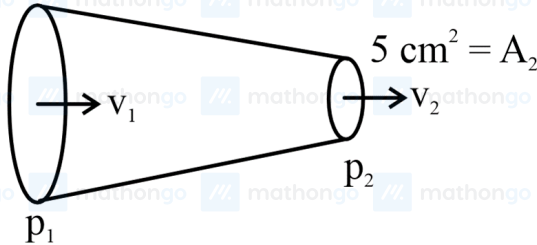
Substitute the value of the known parameter into equation (3) to calculate the required value of the potential for the new drop.

$$V' = 16 \times 10 \text{ mV}$$

$$= 160 \text{ mV}$$

11. (4)

$$A_1 = 10 \text{ cm}^2$$



It is given that

$$A_1 = 10 \text{ cm}^2; A_2 = 5 \text{ cm}^2$$

$$\Delta P = 3 \text{ N m}^{-2}$$

By continuity equation,

$$A_1 v_1 = A_2 v_2$$

$$\Rightarrow v_1 = \frac{A_2 v_2}{A_1} \dots (i)$$

By Bernoulli's equation,

$$\frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 = \Delta P \dots (ii)$$

Using equation (i) in (ii)

$$\Delta P = \frac{\rho}{2} \left(v_2^2 - \left(\frac{A_2 v_2}{A_1} \right)^2 \right)$$

$$\Rightarrow \Delta P = \frac{\rho v_2^2}{2} \left(1 - \left(\frac{A_2}{A_1} \right)^2 \right)$$

$$\Rightarrow 3 = \frac{1.25 \times 10^3 \times v_2^2}{2} \left(1 - \frac{1}{4} \right)$$

$$\Rightarrow v_2^2 = \frac{6 \times 4}{1.25 \times 10^3 \times 3}$$

$$\Rightarrow v_2 = 8 \times 10^{-2} \text{ m s}^{-1}$$

So, the discharge rate is,

$$A_2 v_2 = x \times 10^{-5}$$

$$\Rightarrow x = \frac{5 \times 10^{-4} \times 8 \times 10^{-2}}{10^{-5}} = 4$$

12. (3)

Using Bernoulli's theorem,

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2 \dots (i)$$

Using equation of continuity,

$$A_A V_A = A_B V_B$$

$$\text{So, } 1.5 \times 10^{-4} \times V_A = 25 \times 10^{-6} \times V_B$$

$$V_A = \frac{25 \times 10^{-6}}{1.5 \times 10^{-4}} \times 0.6$$

$$= \frac{1}{10} \text{ m s}^{-1}$$

Substituting in equation (i)

$$\Rightarrow P_A - P_B = \frac{1000}{2} [0.6^2 - 0.1^2] = \frac{1000}{2} \times 0.7 \times 0.5$$

$$= 175 \text{ Pa}$$

13. (3)

The terminal velocity is

$$V_T = \frac{2}{9} r^2 g \frac{(\rho - \rho')}{\eta}$$

$$\Rightarrow V_T \propto r^2$$

Differentiating and dividing by V_T ,

$$\Rightarrow \frac{dV_T}{V_T} = 2 \frac{dr}{r} = 2 \times \frac{0.1}{5}$$

So, the percentage error is

$$\frac{dV_T}{V_T} \times 100 = \frac{0.2}{5} \times 100 = 4\%$$

As can be seen the terminal velocity is directly proportional to the square of the radius. Hence A is true but R is false.

14. (1150)

The pressure inside the bubble is given by

$$\Delta P = P - P_0 = \rho gh + \frac{2T}{r}$$

The data given is

$$r = 10^{-3} \text{ m}$$

$$T = 0.075 \text{ N m}^{-1}$$

$$\rho = 1000 \text{ kg m}^{-3}$$

$$h = 10 \text{ cm}$$

Therefore, the pressure in the bubble is

$$\Delta P = (1000 \times 10 \times 10 \times 10^{-2}) + \frac{2 \times 0.075}{10^{-3}} = 1000 + 150 = 1150 \text{ Pa}$$